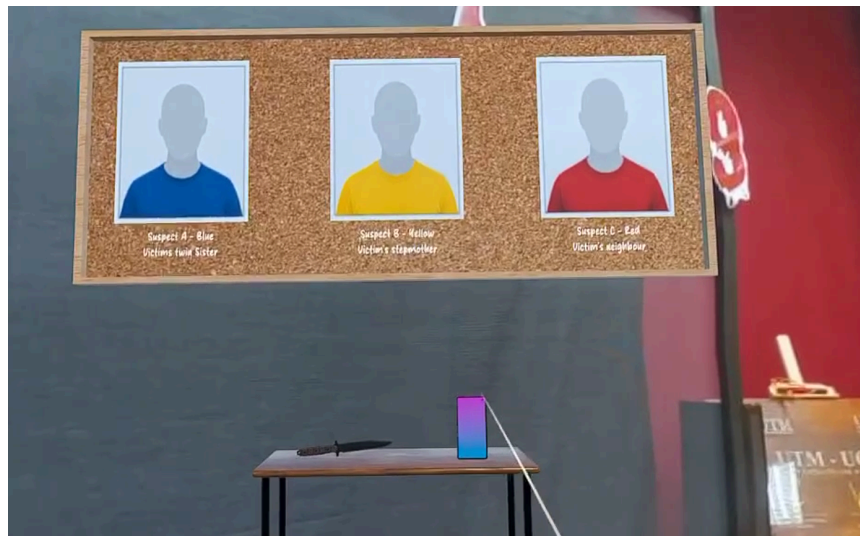


SECV3123 REAL-TIME COMPUTER GRAPHICS
FINAL PROJECT REPORT



HMD + 360 CAMERA + AI

SECTION 01

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1.0 Overview

Beyond The Room VR is a Virtual Reality escape room application developed using Unity and Meta Quest 3. The project combines immersive VR gameplay with real-time 360° environment visualization and AI speech recognition to create an engaging puzzle-solving experience.

The application consists of three different themed rooms, namely Spy Mission, Horror Mystery, and Murder Mystery. Each room presents unique puzzles and objectives that require players to explore the environment, interact with virtual objects, and solve challenges in order to escape successfully.

The project also integrates an Insta360 X4 camera to provide a real-time 360° environment feed and an AI speech recognition feature that allows players to request hints using voice commands. Through the combination of these technologies, the project aims to improve immersion, interaction, and overall user experience in VR escape room applications.

1.1 Team Members and Task Distribution

Table 1: Task Distribution for Beyond The Room VR

Member	Matric No	Responsibility
Wan Nur Fathiya Binti Wan Fauzi	A24CS0311	Development of VR environments, object interaction systems, puzzle mechanisms, player movement, input methods, and AI interaction features within the Unity application.
Kristine Elle Benjamin	A23CS0095	Project planning and management, proposal preparation, report documentation, presentation materials, project coordination, and final report compilation.
Fatin Nadya Rusdyna Binti Rosdi	A24CS0072	Meta Quest 3 and Insta360 X4 setup, usability testing preparation, questionnaire and NASA-TLX analysis, result interpretation, testing documentation, and evaluation report preparation.

1.2 Abstract

This project presents Beyond The Room VR, a Virtual Reality escape room application developed using Unity and Meta Quest 3. The project aims to provide an immersive and interactive puzzle-solving experience by combining VR technology, real-time 360° environment visualization, and AI speech recognition.

The application consists of three escape room themes: Spy Mission, Horror Mystery, and Murder Mystery. Players are required to explore the environment, interact with objects, collect clues, and solve puzzles in order to unlock the exit and complete each room.

To improve immersion, an Insta360 X4 camera was integrated to provide a real-time 360° environment view inside the VR application. In addition, an AI speech recognition feature was implemented to allow players to request hints using voice commands during gameplay.

Usability testing was conducted involving ten participants using Meta Quest 3. The results showed positive feedback in terms of interaction, puzzle design, overall user experience, and ease of use. The findings indicate that the proposed system was able to provide an enjoyable and engaging VR escape room experience.

1.3 Meeting Schedule

Table 2: Meeting Schedule Beyond The Room VR's Team

No	Time/Date	Meeting Platform	Content
1	30 April 2026, Thursday, 4pm - 5pm	Google Meet	Discussed project ideas and selected the VR Escape Room concept. Assigned initial tasks among team members.
2	7 May 2026, Thursday, 8:30pm - 9:30pm	Google Meet	Discussed proposal preparation, project objectives, and system architecture design.
3	12 May 2026, Tuesday, 2:00pm - 3:00pm	Google Meet	Final review and submission of proposal presentation slides.
4	18 May 2026, Monday, 9:00pm - 10:10pm	Google Meet	Reviewed feedback received from the proposal presentation and discussed improvements.

5	20 May 2026, Wednesday, 1:00pm - 1:45pm	Student lounge, N28	Completed proposal revisions and resubmitted the updated proposal.
6	28 May 2026, Tuesday, 3:00pm - 4:00pm	Google Meet	Started VR environment development and integrated basic interactions using Meta Quest 3 controllers.
7	5 June 2026, Friday, 8:00pm - 1:00am	Student Lounge, N28	Developed puzzle mechanisms and implemented gameplay logic for all three escape rooms.
8	10 June 2026, Wednesday, 5:00pm - 7:00pm	Student Lounge, N28	Integrated AI speech recognition and real-time 360° environment visualization.
9	13 June 2026, Saturday, 5:30pm - 11:00pm	Student Lounge, N28	Finalized development stage and submitted Development & Implementation report.
10	20 June 2026, Monday, 8:00pm - 10:00pm	Google Meet	Completed improvements based on feedback and submitted revised development reports.
11	23 June 2026, Tuesday, 5:30pm - 9:00pm	ViCube Lab, N28	Conducted usability testing sessions but failed to complete it.
12	24 June 2026, Wednesday, 12:00pm - 11:30pm	Student Lounge, N28	Conducted usability testing sessions again involving ten participants using Meta Quest 3, repairs project, and recorded application demonstration video and collected final screenshots for documentation. .
13	27 June 2026, Saturday, 7:30pm - 11:45pm	Student Lounge, N28	Analyzed usability testing results and NASA-TLX responses.
14	8 July 2026, Wednesday, 8:00pm - 11:00pm	Student Lounge, N28	Finalized report formatting, presentation slides, and demo reel preparation.
15	10 July 2026, Friday, 12:30pm - 1:00pm	N28a	Final review and submission of complete report, demo reel, and presentation materials.

1.4 Gantt Chart

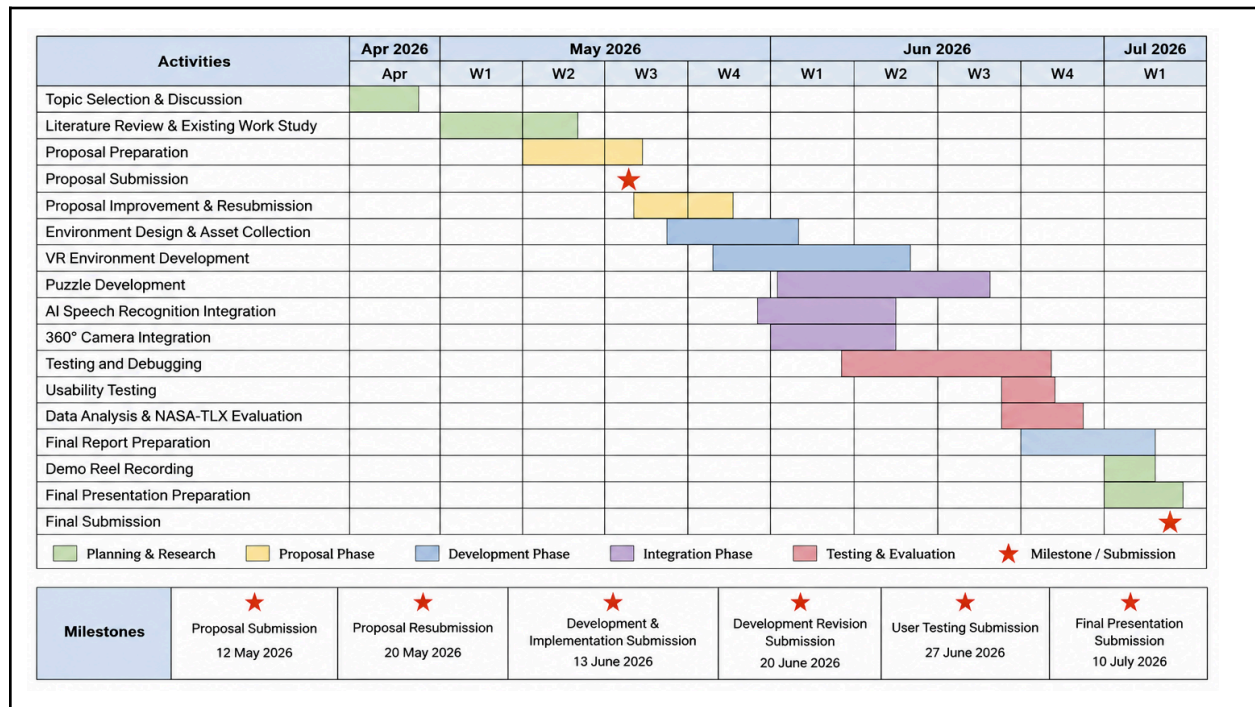


Figure 1: Project development timeline and Gantt chart for Beyond The Room VR.

The project development was carried out progressively following several milestones, including proposal submission, development and implementation, usability testing, and final presentation preparation. The timeline ensured that sufficient time was allocated for system development, feature integration, testing, debugging, and report preparation prior to the final submission.

2.0 Proposal

2.1 Project Description

Beyond The Room VR is a Virtual Reality (VR) escape room application developed using Unity and Meta Quest 3. The project aims to provide an immersive puzzle-solving experience by combining interactive gameplay with real-time 360° environment visualization and AI-based speech recognition.

The application features three different escape room themes, namely Spy Mission, Horror Mystery, and Murder Mystery, each containing unique puzzles and objectives. Players are required to explore the virtual environment, interact with objects, collect clues, and solve puzzles in order to unlock the exit and complete the room.

To enhance immersion, the system integrates an Insta360 camera to provide real-time 360° environmental visualization, allowing users to experience a stronger sense of presence within the virtual world. Additionally, an AI speech recognition module is implemented to provide optional hints when players request assistance using voice commands such as "Give Hint".

The project demonstrates the integration of Virtual Reality technologies, real-time 360° video streaming, object interaction, and AI-assisted gameplay to create an engaging educational and entertainment experience.

2.2 Proposal Slide



Figures 2: Proposal Slide's Content

2.3 Development Platforms

2.3.1 Unity

Unity 6 was used as the primary game development engine for building the virtual environments, implementing game logic, and integrating VR functionalities. The engine provides support for Meta Quest development through XR Interaction Toolkit and Meta XR SDK integration.

2.3.2 Meta Quest 3

Meta Quest 3 served as the main VR hardware platform for user interaction and immersive gameplay experiences. The headset provides inside-out tracking and motion controller support without requiring external sensors.

2.3.3 Insta360 X4

The Insta360 X4 camera was integrated to provide real-time 360° environmental visualization within the VR application, enhancing user immersion and situational awareness.

2.3.4 Meta XR SDK and XR Interaction Toolkit

The Meta XR SDK and XR Interaction Toolkit were used to implement object grabbing, ray interaction, teleportation, and controller input management within the virtual environment.

2.3.5 AI Speech Recognition

An AI speech recognition module was implemented to support voice-based interaction. Players can request gameplay hints using voice commands such as *"Give Hint"* during puzzle-solving activities.

2.3.6 C#

C# was used as the primary programming language for implementing gameplay mechanics, interaction logic, puzzle systems, and speech recognition functionalities.

2.3.7 Visual Studio 2022

Visual Studio 2022 was used as the integrated development environment (IDE) for script development and debugging throughout the project.

3.0 Designing Stage

3.1 Methodology

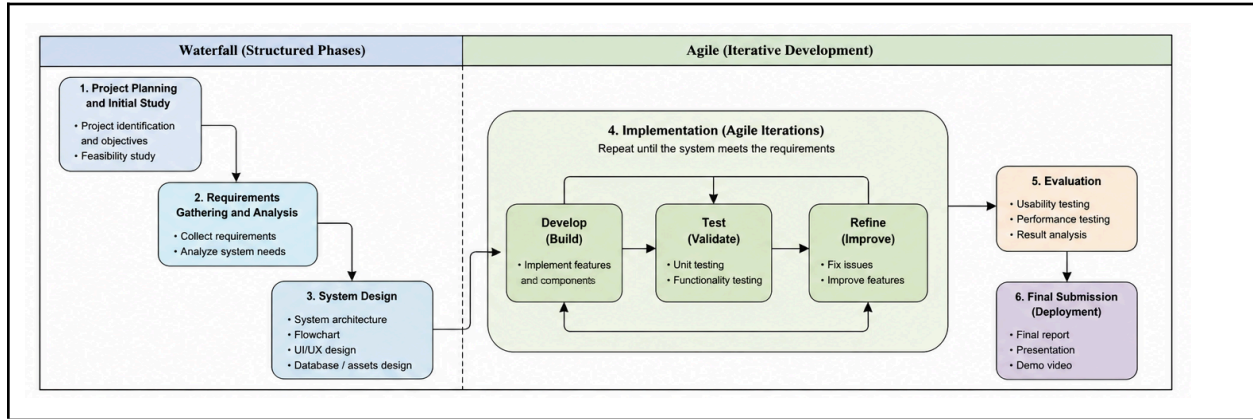


Figure 3: Hybrid Waterfall–Agile methodology of Beyond The Room VR

The development of Beyond The Room VR was carried out using a Hybrid Waterfall–Agile methodology. This approach was selected because the project followed a structured development process based on the course milestones and submission schedule, while still allowing iterative improvements during the implementation stage based on testing results and user feedback.

The overall project structure follows the Waterfall model, which provides a clear sequence from project planning to final evaluation. However, during the implementation phase, Agile practices were adopted to support continuous development, testing, debugging, and feature refinement. This iterative process allowed the team to improve system stability, interaction quality, and overall user experience throughout the development cycle.

The hybrid methodology combines the strengths of both approaches. The Waterfall component ensured that the project remained aligned with the predefined timeline and deliverables, while the Agile component provided flexibility for modifications and improvements identified during development and usability testing.

Table 3 below explains the Hybrid Waterfall–Agile methodology adopted for the development of Beyond The Room VR.

Table 3: Hybrid Waterfall-Agile methodology of Beyond The Room VR

No	Phase	Explanation
1	Project Planning and Initial Study	During this phase, the project objectives, scope, and requirements were identified. Literature review and feasibility studies were also conducted to determine suitable technologies and hardware components for the project.
2	Requirements Gathering and Analysis	The functional and non-functional requirements of the system were collected and analyzed. This included identifying the required VR features, interaction methods, AI speech recognition functionality, and real-time 360° environment visualization requirements.
3	System Design	The overall system architecture, application flow, gameplay mechanics, room designs, and user interaction models were designed before implementation began. The design phase also included planning the puzzle mechanisms and hardware setup.
4	Implementation (Agile Iterations)	The implementation phase followed an iterative Agile approach. Features were developed, tested, improved, and re-tested repeatedly throughout the project. This included the development of VR interactions, puzzle systems, AI speech recognition, and the integration of the Insta360 real-time environment visualization.
5	Evaluation	After all components were integrated, usability testing and performance evaluation were conducted using Meta Quest 3. User feedback and NASA-TLX results were collected and analyzed to identify areas for improvement.
6	Final Submission (Deployment)	The final phase involved preparing the complete report, presentation slides, demonstration video, and other required deliverables for submission and final presentation.

The Hybrid Waterfall-Agile methodology allowed the team to maintain a structured development process while remaining flexible to changes and improvements throughout the implementation stage. This approach was particularly suitable for the development of Beyond

The Room VR, where multiple testing and refinement cycles were required to ensure a stable and enjoyable VR experience for users.

3.2 System Flowchart

The flowchart below shows the overall flow of the application, starting from when the player launches the VR app until they successfully escape the room. It outlines the main steps involved, including initializing the 360° environment, selecting a room theme, exploring and interacting with objects, solving puzzles, and using the AI speech recognition feature if a hint is needed. This flow was used as a general guide during development to ensure that all three escape rooms followed a consistent structure.

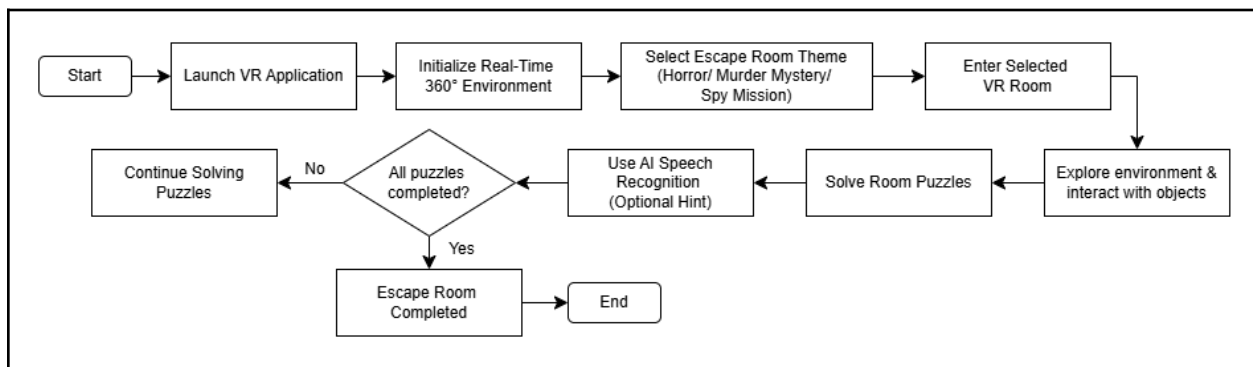


Figure 4: Application Flowchart

3.3 HMD and 360 Camera Setup

This section shows how the hardware devices were arranged during development and testing. The setup consisted of the Meta Quest 3 headset and controllers, a laptop running the Unity application, and the Insta360 X4 camera used to capture the real-time 360° environment. The camera was placed in a fixed position within the room to continuously stream the surrounding environment into the VR application, while the laptop handled the processing and connection between the devices. This arrangement allowed the player to move freely while wearing the headset, with the facilitator monitoring the session from the laptop.

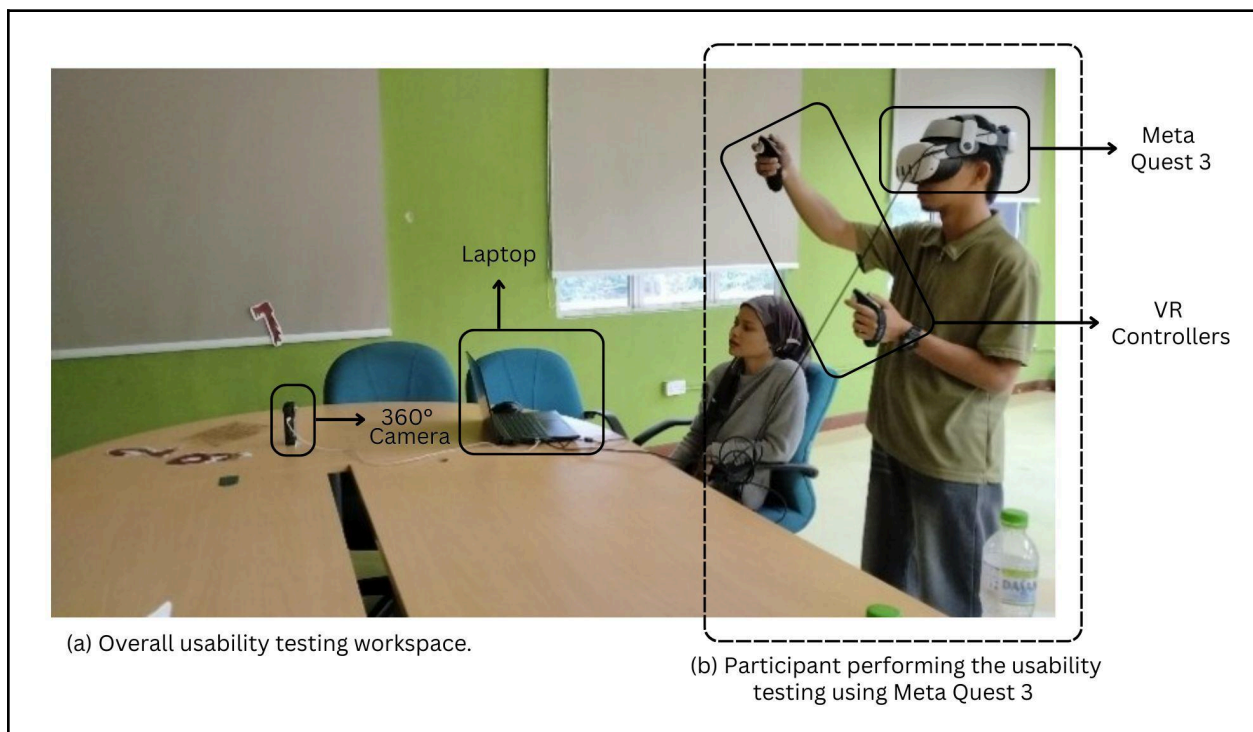


Figure 5: Overall Usability Testing Workspace

3.4 Gameplay Mechanics

The gameplay mechanics were designed separately for each escape room to match their individual storyline and puzzle style, while still following the same overall structure of exploring, solving puzzles, and escaping. As shown in the diagram, all three rooms share a similar core loop: the player explores the environment, interacts with objects to uncover clues, and solves puzzles in order to progress. If the player has difficulty at any point, they can request a hint through the AI speech recognition system, which is explained further in Section 3.4. Once all required puzzles in a room are solved correctly, the exit door unlocks and the player successfully escapes.

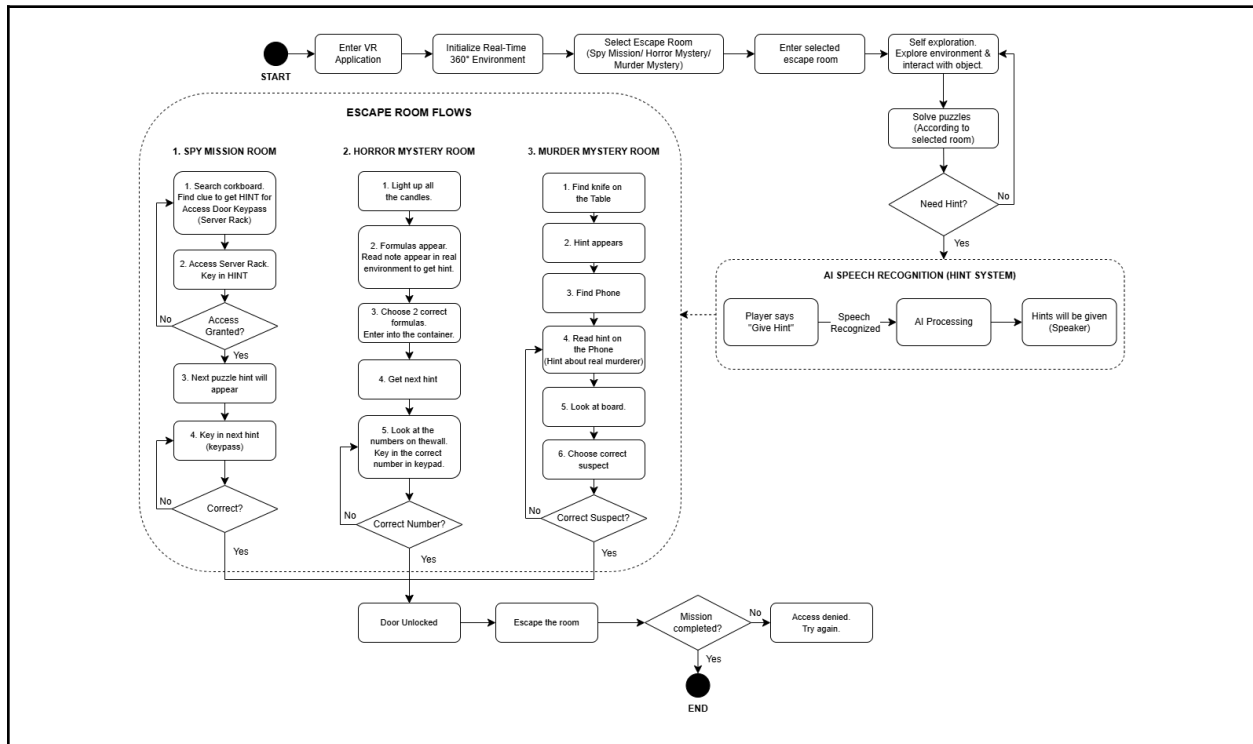


Figure 6: Gameplay Mechanics for Beyond The Room VR

3.5 AI Speech Recognition Design

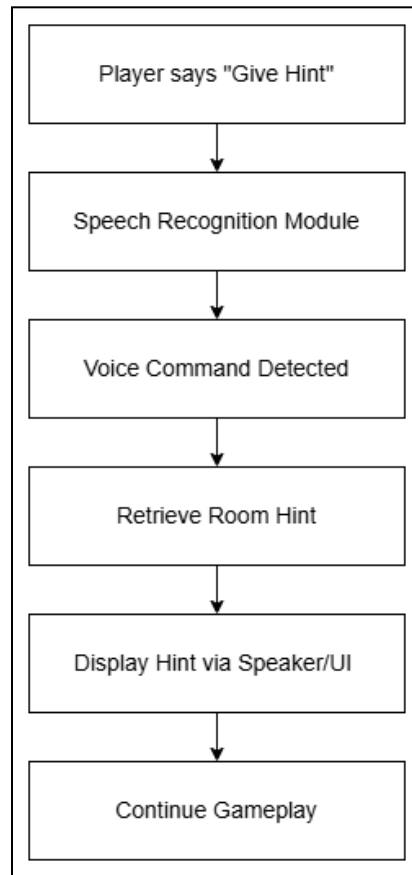


Figure 7: AI Speech Recognition Flow

The AI Speech Recognition feature was implemented to assist players during gameplay by providing hints when needed. Instead of using a dedicated non-player character (NPC), the system allows players to request assistance through voice commands, creating a more natural and immersive interaction experience.

As shown in Figure 7, the process begins when the player says the voice command "Give Hint". The speech recognition module then processes the audio input and detects the command. Once the command is successfully recognized, the system retrieves the corresponding hint based on the player's current room and puzzle progression. The hint is then delivered to the player through the speaker or user interface before the gameplay continues normally.

This feature was designed to reduce player frustration during difficult puzzles while maintaining the challenge and flow of the escape room experience. In addition, the use of voice interaction contributes to the overall immersion by allowing players to communicate naturally with the system without relying on traditional menus or buttons.

4.0 Implementation & Development Stage

4.1 Environment

The environment of Beyond The Room VR was developed using Unity and designed to provide an immersive escape room experience. The application consists of three different room themes, namely Spy Mission Room, Horror Mystery Room, and Murder Mystery Room. Each room contains its own storyline, puzzle sequence, and interaction elements to create different gameplay experiences for the players.

The Spy Mission Room places the player in a locked room where they act as a spy agent who must solve a series of puzzles to obtain the correct passcode and escape. The Horror Mystery Room uses clues placed around the environment and requires players to solve puzzles based on information found within the room. Meanwhile, the Murder Mystery Room allows players to act as detectives by collecting evidence and identifying the correct suspect before escaping the room.

In addition, the project integrates a real-time 360° environment using the Insta360 X4 camera to improve immersion and provide a more realistic VR experience.

4.2 Interactive Elements

Several interactive elements were implemented in the application to improve user engagement and create a more immersive gameplay experience. Players were able to interact with virtual objects using the Meta Quest 3 controllers through the XR Interaction Toolkit provided by Unity.

The implemented interactions include object grabbing, object inspection, keypad interaction, button activation, clue collection, and door unlocking mechanisms. These

interactions allow players to actively participate in puzzle-solving activities rather than simply observing the virtual environment.

The interaction system was designed to be simple and easy to understand so that players with little or no VR experience could still use the application comfortably.

4.3 Puzzles Mechanism

Each escape room contains a different puzzle sequence that matches the room theme and objectives.

In the Spy Mission Room, players were required to search for clues around the room to obtain passwords and access codes. These clues were used to unlock the server rack, workstation, and finally the exit door.

In the Horror Mystery Room, players needed to light up candles and solve formula-based puzzles using hints provided through notes placed in the environment. After obtaining the correct clues, players entered the correct code into the keypad to unlock the door.

For the Murder Mystery Room, players acted as detectives by collecting evidence such as a knife and mobile phone before identifying the correct suspect from the suspect board. Selecting the correct suspect would unlock the exit and complete the room.

The puzzle designs were intended to encourage exploration, observation, and problem-solving skills throughout the gameplay.

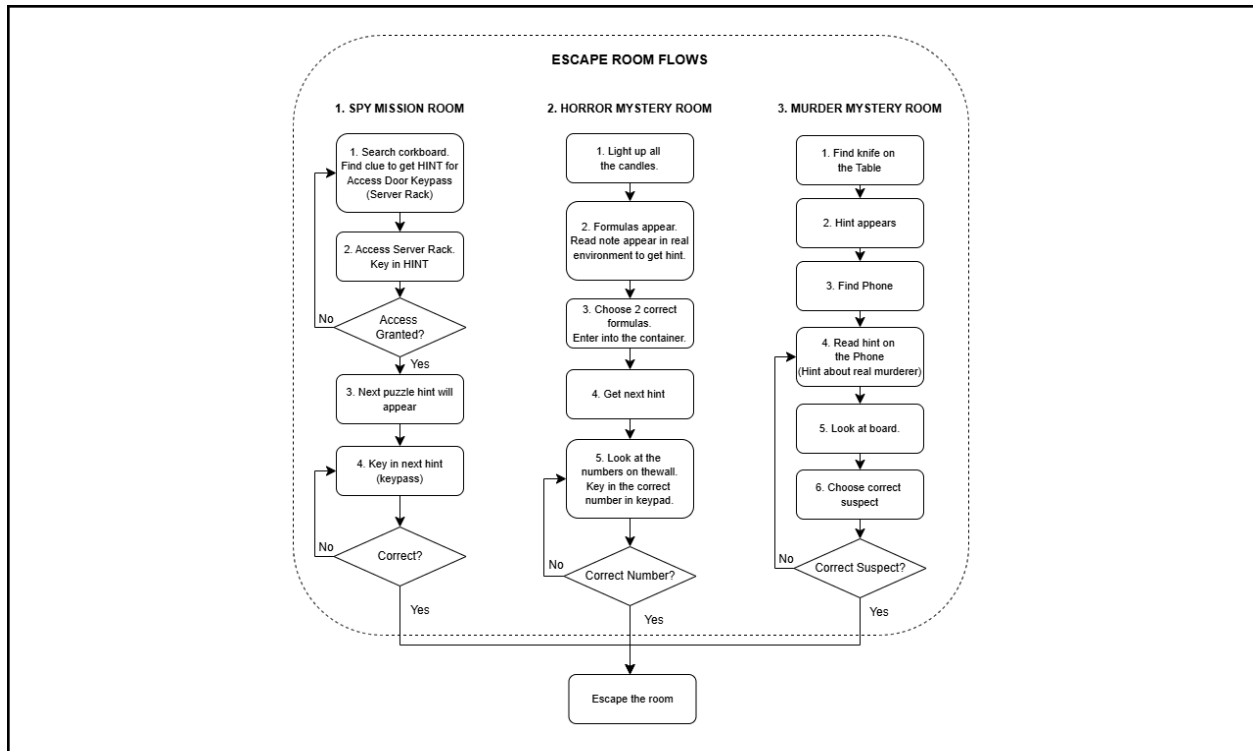


Figure 8: Escape Room Flows

4.4 Player Movement

Player movement within the VR environment was implemented using the locomotion features provided by the Meta XR SDK and XR Interaction Toolkit. Players were able to move around the environment using the Meta Quest 3 controllers while physical head movement was tracked in real time by the headset.

This combination of controller input and head tracking allowed players to explore the environment naturally and interact with objects from different angles. The movement system was designed to provide smooth navigation while maintaining user comfort during gameplay sessions.

4.5 Input Methods

The application supports multiple input methods through the Meta Quest 3 controllers and voice commands. The VR controllers were mainly used for player movement, object interaction, and puzzle solving activities.

The trigger buttons were used for grabbing and interacting with objects, while the thumbsticks were used for navigation within the environment. Ray interaction was also implemented to allow players to select objects that were located further away.

In addition to controller input, voice commands were introduced through the AI speech recognition feature to allow players to request hints during gameplay.

4.6 Settings and Customization

The application was designed with a simple user interface to reduce complexity and improve usability. Players were able to choose their preferred escape room theme from the main menu before starting the gameplay session.

Basic options such as restarting the room and exiting the application were also provided to improve user convenience. Since the main focus of the project was puzzle solving and immersion, only essential customization options were included in the application.

4.7 AI Interaction

An AI speech recognition feature was implemented to provide optional assistance during gameplay. Players could request hints by using the voice command "Give Hint" whenever they encountered difficulties while solving puzzles.

Once the command was detected, the system would identify the player's current room and display a suitable hint related to the current puzzle. This feature was designed to help players continue progressing without completely revealing the puzzle solutions.

The AI interaction system was implemented to improve accessibility and reduce frustration during gameplay while still maintaining the challenge and problem-solving elements of the escape room experience.

5.0 Testing & Evaluation

5.1 Development and Testing Devices

Three laptops were initially considered for the development and testing of Beyond The Room VR. However, performance testing and final evaluation were conducted solely on Laptop A (Acer Nitro V15), as it was the only machine capable of successfully running the application. Laptop B and Laptop C were unable to run the system due to hardware and software incompatibility with the Meta Quest 3 VR development environment and the Unity 6.4 build requirements. As a result, all performance data and usability testing presented in this report were collected using Laptop A only.

Table 4: Comparisons between Devices

	Laptop A	Laptop B	Laptop C
System Model	Acer Nitro V15	Gigabyte Gaming A16 CMH	CMHVictus by HP Gaming Laptop 15-fb0xxx
Ran the application?	Yes	No	No

5.2 Development Device Specifications

The Acer Nitro V15 was the sole machine used for both development and testing throughout the project. Its specifications are summarized in Table 5 below.

Table 5: Device Specifications

Hardware	
System Model	Acer Nitro V15
CPU (Central Processing Unit)	Intel(R) Core(TM) 5 210H
RAM (Random Access Memory)	16GB (15.7 GB usable)
GPU (Graphics Processing Unit)	NVIDIA GeForce RTX 4050
System Type	64-bit operating system, x64-based processor
Software	

Operating Systems	Microsoft Windows 11 Home
Unity Version	Unity 6.4
Integrated	Microsoft Visual Studio 2026
IDE	Microsoft Visual Studio 2026

This configuration provided sufficient processing and graphical performance to support real-time VR rendering, 360° camera streaming, and AI speech recognition processing simultaneously.

5.3 HMD Device Specifications

The Meta Quest 3 was used as the primary head-mounted display (HMD) throughout development and testing. Its key specifications are summarized in the Table 6 below.

Table 6: HMD Device Specifications

Component	Specification
Device	Meta Quest 3
Device Type	Standalone VR
Display Type	2 x LCD binocular
Tracking Type	Hand and body tracking
Controllers	2 x Meta Quest Touch Plus Controllers 6 DoF
Connectivity	USB Type-C
Refresh Rate	120 Hz

5.4 360 Camera Device Specifications

The Insta360 X4 camera was used to capture and stream the real-world environment into the VR application in real time. Its role and specifications are summarized in Table 7 below.

Table 7: 360 Camera Device Specifications

Component	Specification
Device	Insta360 X4
Video Resolution	3840x1920@60/50/30/25/24fps
Video Format	INSV
Connectivity	USB Type-C live streaming to development laptop

Figure 9 below shows how the hardware was used during the testing of the system.

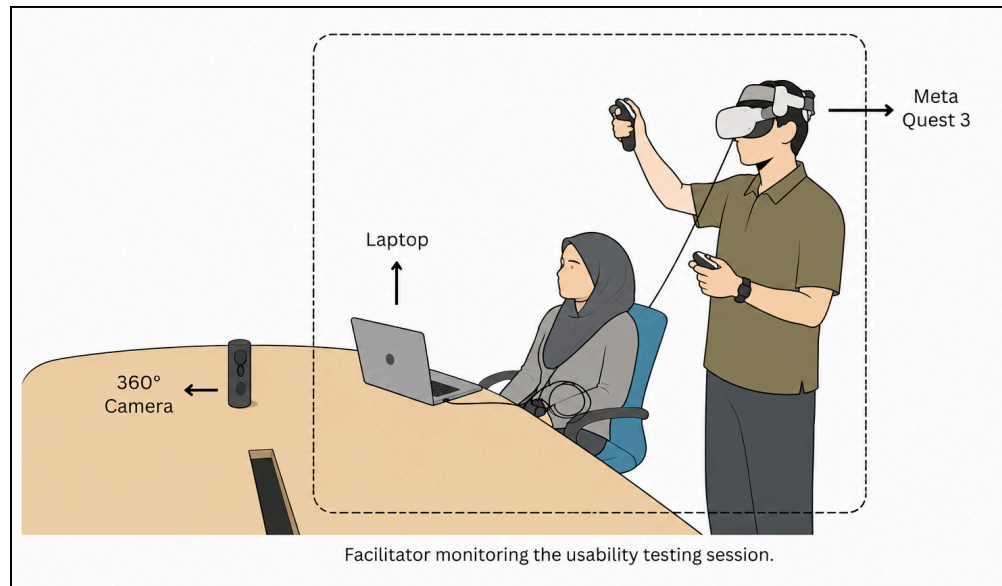


Figure 9: Hardware components used during testing

5.5 User Testing Methodology

Usability testing was conducted offline at the ViCube Lab, N28, UTM, involving a total of 10 participants. The methodology and testing details are summarized in the Table 8 below.

Table 8: Methodology and Testing details

Aspect	Details
Number of Participants	10
Gender Distribution	70% Male, 30% Female
Age Range	20-25
Prior VR Experience	90% had used VR before, 10% had not
Testing Procedure	Participants were briefed on the session objectives before wearing the Meta Quest 3 headset and receiving a brief controller introduction. They then explored the VR environment, interacted with objects, and solved puzzles while facilitators recorded observations. Upon completion, participants filled out a post-test questionnaire and NASA-TLX survey via Google Forms.
Duration of Each Session	Approximately 5-10 minutes, depending on the participant's gameplay speed.
Testing Procedure	Participants were briefed on session objectives, given a short controller introduction, then explored the VR environment, interacted with objects, and solved puzzles while facilitators observed. Afterward, participants completed a Likert-7 questionnaire and NASA-TLX survey via Google Forms.
Tools & Equipments	Meta Quest 3 headset and controllers, development laptop running the Unity application, Insta360 X4 camera, and Google Forms for data collection.
Executable Package for Participants	Not Applicable (Application was pre-installed on Meta Quest 3).
Link for Questionnaire	GoogleForm_Usability_Testing_and_NASA-TLX
Results and Materials	GoogleDrive_Link_to_Results_and_Materials

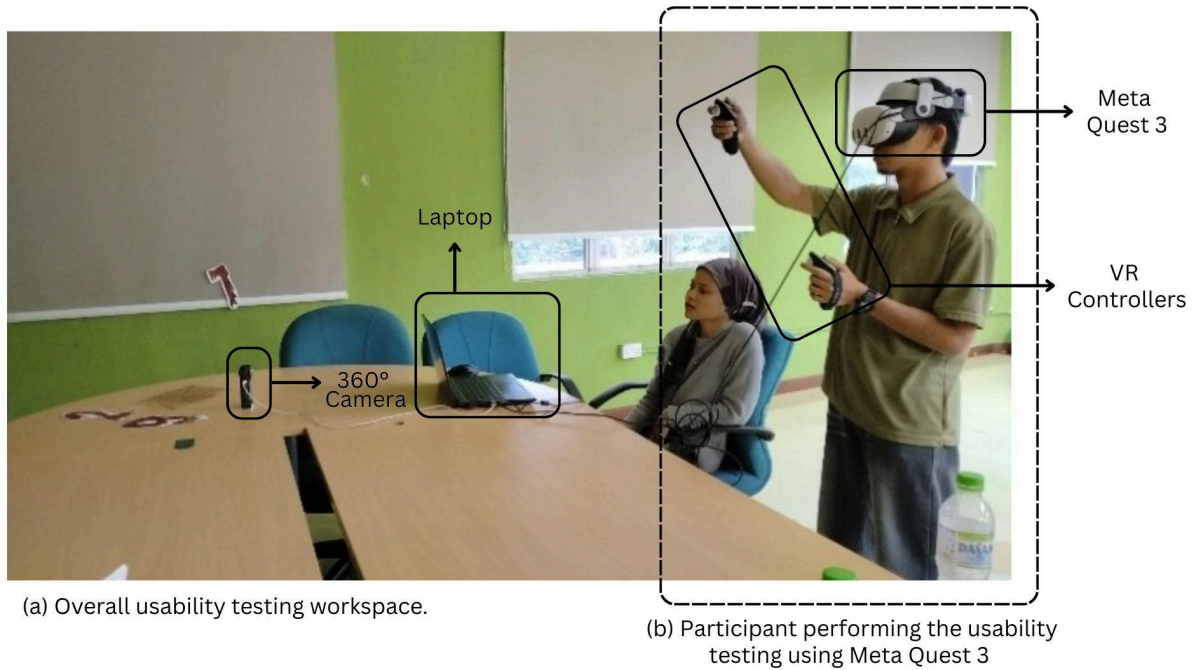


Figure 10: Overall usability testing workspace (left) and participant performing the test photo (right)

5.6 Result Analysis

5.6.1 Performance Testing

Performance testing showed that the application maintained a stable frame rate close to the Meta Quest 3's native 72Hz target across the Menu, Escape Room Idle, and Escape Room Puzzle scenarios. However, frame rate dropped sharply during Room Transitions, likely due to simultaneous asset loading blocking the main thread.

Table 9: Performance Testing Result

Metric	Menu	Escape Room Idle	Escape Room Puzzle	Room Transition
FPS	71.9	72.0	72.1	1.8
Tris	10.3k	14.0k	26.8k	28.2k
Verts	15.8k	19.7k	40.4k	31.1k

5.6.2 User Experience Evaluation

Across all categories, the majority of responses fell within the 4–5 range, indicating generally positive reception. Table 10 below summarizes the average score for each statement.

Table 10: User Experience Evaluation Results

Category	Statement	Mean Score (out of 5)
VR Environment	The virtual environment was immersive and realistic.	4.3
	The real-time 360° camera view enhanced my VR experience.	4.3
	The visual quality of the environment was satisfactory.	4.4
Interaction & Controls	The VR controls were easy to understand and use.	4.6
	Grabbing and interacting with objects felt natural.	4.5
	Navigation within the VR environment was smooth.	4.4
Puzzle Design	The puzzles were interesting and engaging.	4.5
	The clues provided were helpful and understandable.	4.4
	The difficulty level of the puzzles was appropriate.	4.4
AI Speech Recognition	The speech recognition feature responded correctly to my commands.	3.8
	The AI hint system was helpful during gameplay	3.8
Overall Experience	The application was easy to use.	4.5
	The VR escape room	4.6

	experience increased my interest in VR technology.	
	I would recommend this application to others.	4.6

The results show that Interaction & Controls and Overall Experience received the highest ratings, confirming that the VR movement, object grabbing, and navigation systems were well received by users. In contrast, the AI Speech Recognition category received the lowest average score (3.8/5), with 50% of participants rating it only a 3 out of 5 for both command accuracy and hint helpfulness, thus this evaluation is consistent with the qualitative feedback requesting improvements to the AI voice assistant.

5.6.3 NASA Task Load Index

Table 11 shows the result of NASA Task Load Index, filled out by 10 participants testing the system.

Table 11: NASA Task Load Index Result

Dimension	Mean Score (out of 10)	Interpretation
Mental Demand	4.9	Moderate mental effort required
Physical Demand	3.7	Relatively low physical strain
Temporal Demand (Rushed/Hurried)	3.3	Participants generally did not feel rushed
Performance (Success)	7.9	Participants felt largely successful in completing tasks
Effort	3.9	Moderate effort required to accomplish tasks
Frustration	3.3	Low to moderate frustration levels

The NASA-TLX results indicate a favorable overall workload profile: participants reported a high sense of task success (7.9/10) alongside relatively low frustration (3.3/10) and physical

demand (3.7/10). Mental demand was moderate (4.9/10), which is reasonable given the puzzle-solving nature of the escape rooms.

5.6.4 Room Preference and Open-Ended Feedback

When asked which room they enjoyed most, participants were evenly split between the Murder Mystery Room (50%) and the Spy Mission Room (50%), with no participants selecting the Horror Room. Common reasons given included the puzzles being "easy and fun to complete," clear hints, and rooms that felt "more realistic" or offered "more interaction than other rooms."

For open-ended feedback, participants most frequently praised the smoothness of the grabbing/interaction system, the puzzle and mystery-solving elements, and the variety of escape room themes available. The most common suggestions for improvement centered on the AI voice assistant, with several participants specifically requesting better AI voice command recognition, along with suggestions to add sound cues on interactions and increase puzzle difficulty for a more challenging experience.

6.0 Conclusion

Beyond The Room VR successfully demonstrates the integration of Virtual Reality technology, real-time 360° environmental visualization, and AI-based speech recognition into a single, cohesive escape room experience. Developed using Unity and the Meta Quest 3, the application delivers three distinct themed rooms, which are Spy Mission, Horror Mystery, and Murder Mystery. Each offering unique puzzles, interaction mechanics, and narrative elements designed to encourage exploration, observation, and problem-solving.

Throughout the development process, the project achieved its core objectives. The Meta XR SDK and XR Interaction Toolkit enabled natural object grabbing, ray interaction, and locomotion, while the Insta360 X4 camera was successfully integrated to stream a real-time 360° view of the physical environment into the VR application, enhancing player immersion. The AI speech recognition module allowed players to request contextual hints through simple voice commands, reducing frustration without fully revealing puzzle solutions.

Performance testing confirmed that the application maintained a stable frame rate close to the Meta Quest 3's native refresh target during normal gameplay, though a significant frame rate drop during room transitions was identified as a technical limitation likely caused by simultaneous asset loading. Usability testing involving ten participants further validated the project's design, with particularly strong feedback for interaction and controls, puzzle design, and overall user experience. Participants also reported a high sense of task success and relatively low frustration on the NASA-TLX evaluation, indicating that the application was both engaging and manageable to use even for those with limited prior VR experience. The AI speech recognition feature, while functional, was consistently identified as the area most in need of improvement, particularly regarding command recognition accuracy and response consistency.

Overall, Beyond The Room VR met its objective of creating an immersive, AI-assisted VR escape room experience that combines interactive gameplay with real-world environmental visualization. The positive usability results affirm the viability of combining HMD-based VR, live 360° camera feeds, and AI voice interaction within a single application. Moving forward, future development should prioritize optimizing asset loading during room transitions to eliminate the observed frame rate drop, improving the accuracy and responsiveness of the AI speech recognition system, and resolving the real-time 360° camera display issue reported by some participants. With these refinements, the application has strong potential to be further expanded with additional room themes, enhanced puzzle complexity, and improved voice interaction to deliver an even more polished and immersive escape room experience.

7.0 References

EVERYDAY VR Gameplay (2024, February 10). *AVO Escape Space | FULL PLAYTHROUGH |*

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APPENDIX A

User Manual

A.1 System Requirements

Hardware	Software
• Laptop/PC capable of running Unity VR application • Meta Quest HMD • Hand controllers • 360° camera	• Beyond the Room VR Unity project • Meta Quest Link / Air Link • OBS Studio for webcam video capture • Windows microphone permission enabled

A.2 Application Setup

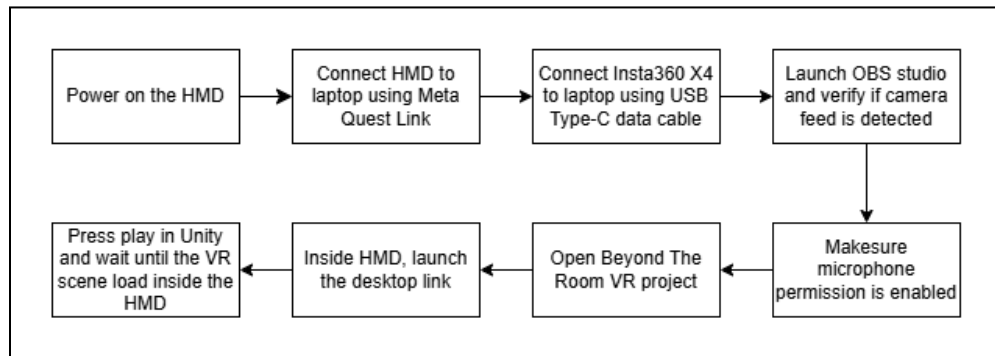


Figure 11: Flowchart of how to setup Beyond the Room VR application

A.3 Starting the VR Escape Room

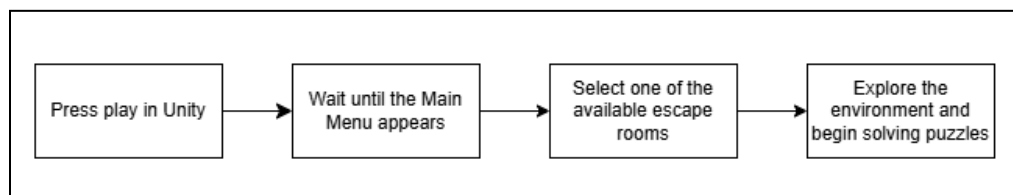


Figure 12: Flowchart of Starting the VR Escape Room

A.4 User Controls

Control	Action
Left Thumbstick	Move forward, backward, left and right
Right Thumbstick	Snap turn left and right
Grab Button	Select UI
Trigger Button	Grab object
Voice Command	Request Hint

A.5 Puzzle Gameplay Guide

A.5.1 Spy Mission

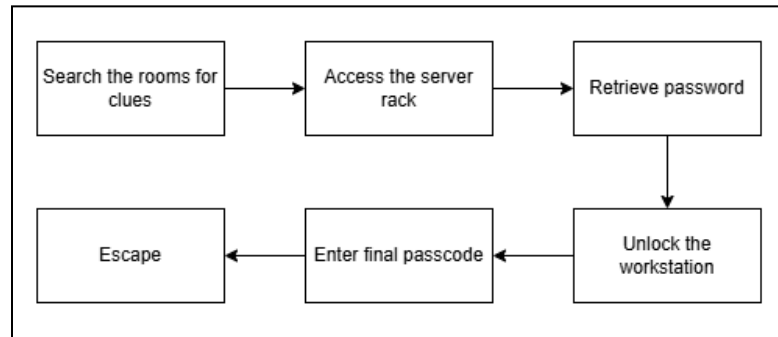


Figure 13: Puzzle Gameplay Flowchart for Spy Mission

A.5.2 Horror Mystery

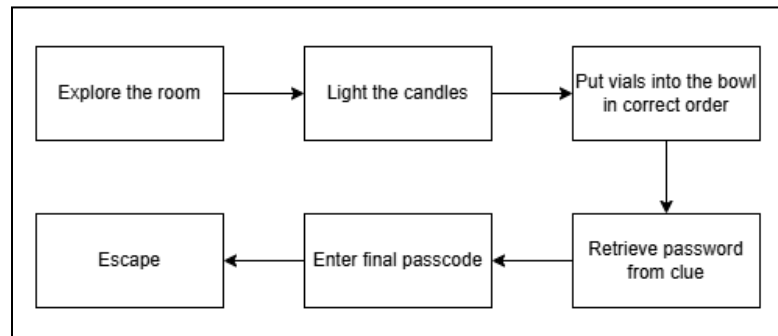


Figure 14: Puzzle Gameplay Flowchart for Horror Mystery

A.5.3 Murder Mystery

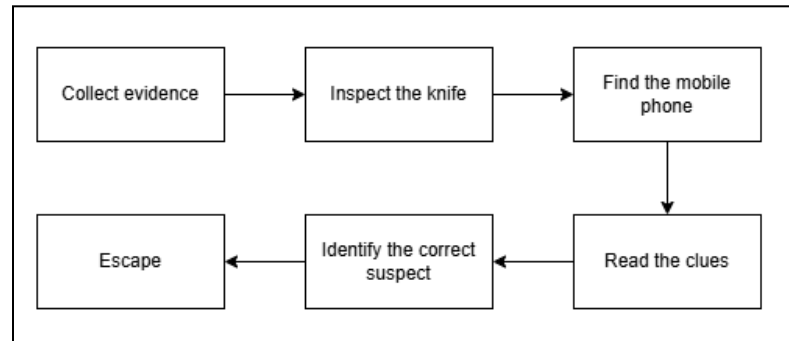


Figure 15: Puzzle Gameplay Flowchart for Murder Mystery

A.6 AI Voice Commands

Voice Command	Function
“Give Hint”	Provides a hint related to the player’s current puzzle.

A.7 Troubleshooting

Problem	Solution
Meta Quest not detected	- Restart Meta Quest Link or reconnect the USB cable.
No camera feed	- Verify the Insta360 X4 is connected and selected correctly in OBS Studio. - Make sure the USB Type-C is a data cable.
Voice command not working	- Ensure the microphone is enabled in Windows and that the environment is quiet.
Black screen in headset	- Restart the Unity application and reconnect the headset.